AFRILANG Stdlib

std/c/mlsoftmax

Français. Bibliothèque AFRILANG ‘std/c/mlsoftmax‘ (niveau complex, catégorie complex). Elle expose 6 fonction(s) : softmax, crossEntropy, argmax, oneHot, logSoftmax, entropy.
‘import ”std/c/mlsoftmax”‘ · ‘use mlsoftmax‘
- ‘softmax(logits list number) → list number‘ - ‘crossEntropy(probs list number, target number) → number‘ - ‘argmax(v list number) → number‘ - ‘oneHot(idx number, size number) → list number‘ - ‘logSoftmax(logits list number) → list number‘ - ‘entropy(probs list number) → number‘

English. AFRILANG library ‘std/c/mlsoftmax‘ (tier complex, category complex). Exports 6 function(s): softmax, crossEntropy, argmax, oneHot, logSoftmax, entropy.
‘import ”std/c/mlsoftmax”‘ · ‘use mlsoftmax‘
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Catégorie / Category	Modules complex (c/)
Niveau / Tier	complex
Fonctions / Functions	6

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Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/mlsoftmax' (niveau complex, catégorie complex). Elle expose 6 fonction(s) : softmax, crossEntropy, argmax, oneHot, logSoftmax, entropy.
'import "std/c/mlsoftmax"' · 'use mlsoftmax'

```
- `softmax(logits list number) → list number`
- `crossEntropy(probs list number, target number) → number`
- `argmax(v list number) → number`
- `oneHot(idx number, size number) → list number`
- `logSoftmax(logits list number) → list number`
- `entropy(probs list number) → number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/mlsoftmax"
use mlsoftmax
```

```
Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés — graphes, NLP, réseaux complexes.
```

3. Référence API

```
•
 softmax(logits list number) → list•
 crossEntropy(probs list number, target number) → number•
 argmax(v list number) → number•
 oneHot(idx number, size number) → list•
 logSoftmax(logits list number) → list•
 entropy(probs list number) → number
```

4. Exemples d'utilisation

```
import "std/c/mlsoftmax"
use mlsoftmax
```

```
create result = softmax(/* args */)
say result
```

```
create result = crossEntropy(/* args */)
say result
```

```
create result = argmax(/* args */)
say result
```

```
create result = oneHot(/* args */)
say result
```

```
create result = logSoftmax(/* args */)
say result
```

```
create result = entropy(/* args */)
say result
```

5. Bonnes pratiques

```
•
 Préférez `import "std/c/mlsoftmax"` en tête de fichier.•
 Lancez `afrilang check` avant de livrer.•
 Combinez avec les modules stdlib de la même catégorie.•
 Voir /stdlib/ pour le source live et les démos playground.•
 Signalez les problèmes sur GitHub si une export manque.
```

6. Annexe — source

module mlsoftmax

```
function softmax(logits list number) returns list number
end
```

```
function crossEntropy(probs list number, target number) returns number
end
```

```
function argmax(v list number) returns number
end
```

```
function oneHot(idx number, size number) returns list number
end
```

```
function logSoftmax(logits list number) returns list number
end
```

```
function entropy(probs list number) returns number
end
end
```

English

1. Overview

AFRILANG library ‘std/c/mlsoftmax‘ (tier complex, category complex). Exports 6 function(s): softmax, crossEntropy, argmax, oneHot, logSoftmax, entropy.

‘import ”std/c/mlsoftmax”‘ · ‘use mlsoftmax‘

```
- `softmax(logits list number) → list number`
- `crossEntropy(probs list number, target number) → number`
- `argmax(v list number) → number`
- `oneHot(idx number, size number) → list number`
- `logSoftmax(logits list number) → list number`
- `entropy(probs list number) → number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/mlsoftmax"
use mlsoftmax
```

```
Tier: complex · Category: Complex modules (c/)
Advanced domains — graphs, NLP, complex networks.
```

3. API reference

```
•
 softmax(logits list number) → list•
 crossEntropy(probs list number, target number) → number•
 argmax(v list number) → number•
 oneHot(idx number, size number) → list•
 logSoftmax(logits list number) → list•
 entropy(probs list number) → number
```

4. Usage examples

```
import "std/c/mlsoftmax"
use mlsoftmax
```

```
create result = softmax(/* args */)
say result
```

```
create result = crossEntropy(/* args */)
say result
```

```
create result = argmax(/* args */)
say result
```

```
create result = oneHot(/* args */)
say result
```

```
create result = logSoftmax(/* args */)
say result
```

```
create result = entropy(/* args */)
say result
```

5. Best practices

```
•
 Prefer `import "std/c/mlsoftmax"` at the top of your file.•
 Run `afrilang check` before shipping.•
 Combine with related stdlib modules from the same category.•
 See /stdlib/ for the live source and playground demos.•
 Report issues on GitHub if an export is missing or undocumented.
```

6. Appendix — source

module mlsoftmax

```
function softmax(logits list number) returns list number
end
```

```
function crossEntropy(probs list number, target number) returns number
end
```

```
function argmax(v list number) returns number
end
```

```
function oneHot(idx number, size number) returns list number
end
```

```
function logSoftmax(logits list number) returns list number
end
```

```
function entropy(probs list number) returns number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/mlsoftmax"`
- Vérification : `afrilang check`
- Documentation : https://afrilang.dev/stdlib/std/c/mlsoftmax/

Source (extrait)

```
module mlsoftmax

function softmax(logits list number) returns list number
end

function crossEntropy(probs list number, target number) returns number
end

function argmax(v list number) returns number
end

function oneHot(idx number, size number) returns list number
end

function logSoftmax(logits list number) returns list number
```

```
end

function entropy(probs list number) returns number
end
end
```